

Auteur: Yoren Collier
 Studierichting: Informatica
 Oefeningenreeks 5A nr. 6.18

$$BI \int_0^1 \frac{x^2 + \alpha x + \alpha}{x^3 + 1} dx \Rightarrow OBI = \int \frac{x^2 + \alpha x + \alpha}{x^3 + 1} dx$$

$$= \int \frac{x^2 + \alpha(x+1)}{(x+1)(x^2 - x + 1)}$$

We nemen $b(x)$ gelijk aan de breuk na de integraal.

$$b(x) = \frac{A}{(x+1)} + \frac{Bx+C}{(x^2-x+1)}$$

$$A = \frac{x^2 + \alpha(x+1)}{(x^2 - x + 1)} \Big|_{x=-1} = \frac{(-1)^2 + \alpha(-1+1)}{(-1)^2 - (-1) + 1} = \frac{1}{3}$$

$$x^2 - x + 1 = 0$$

$$\Delta = (-1)^2 - 4 \cdot 1 \cdot 1 = -3$$

$$x^+ = \frac{1 + \sqrt{-3}}{2} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$x^- = \frac{1 - \sqrt{-3}}{2} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$B \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) + C = \frac{x^2 + \alpha(x+1)}{(x+1)} \Big|_{x=\frac{1}{2} + \frac{\sqrt{3}}{2}i}$$

$$= \frac{\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)^2 + \alpha \left(\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) + 1 \right)}{\left(\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) + 1 \right)} = \frac{\frac{-1}{2} + \frac{\sqrt{3}}{2}i + \alpha \left(\frac{3}{2} + \frac{\sqrt{3}}{2}i \right)}{\frac{3}{2} + \frac{\sqrt{3}}{2}i}$$

$$= \frac{\frac{1}{2}(-1 + \sqrt{3}i) + \alpha \cdot \frac{1}{2}(3 + \sqrt{3}i)}{\frac{1}{2}(3 + \sqrt{3}i)} = \frac{[(-1 + \sqrt{3}i) + \alpha(3 + \sqrt{3}i)](3 - \sqrt{3}i)}{(3 + \sqrt{3}i)(3 - \sqrt{3}i)} =$$

$$\frac{4\sqrt{3}i + \alpha \cdot (12)}{12} = \frac{\sqrt{3}i + 3\alpha}{3} = \alpha + \frac{\sqrt{3}}{3}i$$

$$\frac{1}{2}B + \frac{\sqrt{3}}{2}iB + C = \alpha + \frac{\sqrt{3}}{3}i$$

$$\begin{cases} \frac{1}{2}B + C = \alpha \\ \frac{\sqrt{3}}{3}iB = \frac{\sqrt{3}}{3}i \end{cases} \Leftrightarrow \begin{cases} \frac{1}{2} + C = \alpha \\ B = 1 \end{cases} \Leftrightarrow \begin{cases} C = \alpha - \frac{1}{2} \\ B = 1 \end{cases}$$

$$I = \frac{1}{3} \int \frac{1}{x+1} dx + \int \frac{x + \alpha - \frac{1}{2}}{x^2 - x + 1} dx$$

$$\begin{aligned}
D(x^2 - x + 1) &= 2x - 1 \\
&= \frac{1}{3} \ln|x+1| + c + \frac{1}{2} \int \frac{2x + 2\alpha - 1}{x^2 - x + 1} dx = \frac{1}{3} \ln|x+1| + c + \frac{1}{2} \left[\int \frac{1}{x^2 - x + 1} (2x - 1) dx + \int \frac{2\alpha}{x^2 - x + 1} dx \right] \\
\text{Stel: } t = x^2 - x + 1 &\Rightarrow dt = (2x - 1) dx \\
\frac{1}{3} \ln|x+1| + c + \frac{1}{2} &\left[\int \frac{1}{t} dt + 2\alpha \int \frac{1}{x^2 - x + \frac{1}{4} + \frac{3}{4}} dx \right] \\
&= \frac{1}{3} \ln|x+1| + c + \frac{1}{2} \left[\ln|x^2 - x + 1| + c + 2\alpha \int \frac{1}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}} dx \right] \\
\text{Stel: } y = x - \frac{1}{2} &\Rightarrow dy = dx \\
&= \frac{1}{3} \ln|x+1| + c + \frac{1}{2} \left[\ln|x^2 - x + 1| + c + 2\alpha \int \frac{1}{y^2 + \frac{3}{4}} dy \right] \\
&= \frac{1}{3} \ln|x+1| + c + \frac{1}{2} \left[\ln|x^2 - x + 1| + c + 2\alpha \cdot \frac{1}{\frac{\sqrt{3}}{2}} \text{Bgtan} \left(\frac{\frac{y}{\frac{\sqrt{3}}{2}}}{\frac{2}{2}} \right) + c \right] = \\
&\frac{1}{3} \ln|x+1| + c + \frac{1}{2} \ln|x^2 - x + 1| + c + 2\alpha \cdot \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \text{Bgtan} \left(\frac{2 \left(x - \frac{1}{2}\right)}{\sqrt{3}} \right) + c = \\
&\frac{1}{3} \ln|x+1| + \frac{1}{2} \ln|x^2 - x + 1| + \frac{2\alpha}{\sqrt{3}} \text{Bgtan} \left(\frac{2x - 1}{\sqrt{3}} \right) + c \\
BI &= \left[\frac{1}{3} \ln|x+1| + \frac{1}{2} \ln|x^2 - x + 1| + \frac{2\alpha}{\sqrt{3}} \text{Bgtan} \left(\frac{2x - 1}{\sqrt{3}} \right) \right]_0^1 \\
&= \frac{1}{3} \ln|2| + \frac{1}{2} \ln|1| + \frac{2\alpha}{\sqrt{3}} \text{Bgtan} \left(\frac{1}{\sqrt{3}} \right) - \left(\frac{1}{3} \ln|1| + \frac{1}{2} \ln|1| + \frac{2\alpha}{\sqrt{3}} \text{Bgtan} \left(\frac{-1}{\sqrt{3}} \right) \right) \\
&= \frac{1}{3} \ln|2| + 0 + \frac{2\alpha}{\sqrt{3}} \cdot \frac{\pi}{6} - \frac{1}{3} \cdot 0 - \frac{1}{2} \cdot 0 - \frac{2\alpha}{\sqrt{3}} \cdot \frac{-\pi}{6} = \frac{1}{3} \ln|2| + \frac{2\alpha\pi}{6\sqrt{3}} - \frac{-2\alpha\pi}{6\sqrt{3}} = \\
&\frac{1}{3} \ln|2| + \frac{2\alpha\pi}{3\sqrt{3}} = \frac{1}{3} \ln|2| + \frac{2\alpha\pi}{3\sqrt{3}}
\end{aligned}$$

References